

MUNTAKIM RAHAMAN

Prospective PhD Applicant - AI-Enabled Cyber-Physical Systems Security, Anomaly Detection and Privacy Preservation

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APPLICATION FIT SNAPSHOT

Target area	Securing AI-enabled cyber-physical systems through cybersecurity, anomaly detection, privacy-preserving monitoring, robust machine learning and reproducible evaluation.
Eligibility match	B.Sc. in Computer Science and Engineering with CGPA 3.74/4.00; Master of Engineering Science by Research in Cybersecurity and Networking, expected February 2027.
Publication strength	Published first-author journal article in adaptive encrypted traffic classification; accepted IEEE TENSYP paper; additional manuscripts under review or ready for submission.
Technical fit	Machine learning, deep learning, few-shot/meta-learning, zero-day detection, drift robustness, IoT/IoMT traffic security, O-RAN security, encrypted traffic analysis and privacy-aware network monitoring.
Implementation strength	Five-plus years of applied AI/data-science experience plus research software in Python, Rust, FastAPI, React, Docker and reproducible benchmarking pipelines.

RESEARCH PROFILE

Graduate Research Assistant and Master's-by-research student in Cybersecurity and Networking at Multimedia University, Malaysia. My current research sits at the intersection of AI-assisted cybersecurity, privacy-preserving network monitoring and adaptive defence for networked cyber-physical infrastructure. I develop reproducible machine-learning frameworks for encrypted traffic classification, zero-day and anomaly detection, domain-shift robustness and drift recovery, with a complementary focus on quantum-safe and crypto-agile security for future communication systems. I also bring industry experience as a data scientist building anomaly detection, predictive-maintenance, time-series, geospatial analytics and deployable AI systems.

FIT TO DEAKIN PHD TOPIC

- AI-enabled CPS cybersecurity: experience designing ML/DL security pipelines for IoT, IoMT, O-RAN and encrypted network environments where operational data are dynamic, noisy and partially labelled.
- Anomaly and zero-day detection: research on few-shot/meta-learning, open-world traffic classification, adaptation-detection trade-offs, domain shift and drift recovery for novel or evolving threats.
- Privacy preservation: focus on payload-free and leakage-aware encrypted traffic intelligence using metadata such as packet bursts, flow statistics, TLS/QUIC behaviour and protocol-level signals.
- Research software and benchmarking: built reproducible testbeds with seed-level reporting, same-budget baselines, ablations, dataset export and controller interfaces for rigorous PhD-level experimentation.
- Cyber-physical relevance: applied AI background includes industrial predictive maintenance, time-series anomaly detection, sensor/plant data analytics and deployed stakeholder-facing systems.

EDUCATION

Master of Engineering Science by Research in Cybersecurity and Networking | *Multimedia University, Malaysia* | Graduate Research Assistant | 2025 - expected February 2027

- Research focus: AI-assisted threat detection, encrypted traffic classification, privacy-aware network monitoring, QKD/PQC orchestration, adaptive network control and quantum-safe service management.
- Selected outputs: TrafficMAML framework; QKD/PQC service-management simulator; quantum-safe O-RAN crypto-agility research; systematic review on adaptive encrypted traffic classification.

Bachelor of Science in Computer Science and Engineering | *Bangladesh Army International University of Science and Technology* | 2016 - 2020 | CGPA: 3.74/4.00

- Relevant coursework: Cryptography and Network Security, Computer Networks, Distributed Systems, Machine Learning and Software Engineering.

RESEARCH EXPERIENCE

Graduate Research Assistant | *Multimedia University, Malaysia* | 2025 - Present

- Conduct research on AI-assisted cybersecurity, encrypted traffic intelligence, quantum-safe networking, crypto-agility and adaptive security for O-RAN.
- Develop reproducible evaluation protocols for few-shot encrypted traffic classification, zero-day detection, domain-shift robustness and drift recovery.
- Build research software pipelines for experiment automation, seed-level reproducibility, baseline controls, ablation studies and dataset export.
- Prepare manuscripts for high-impact venues including IEEE TNSM, IEEE Access, IEEE WIFS and Elsevier Computer Networks.

SELECTED RESEARCH SOFTWARE AND PROJECTS

TrafficMAML Framework | *Few-shot and privacy-aware encrypted traffic intelligence* | [GitHub](#):

github.com/muntakim1/trafficmaml-framework

A reproducible few-shot learning framework for encrypted traffic classification under domain shift, zero-day classes and drift conditions.

- Implements few-shot/meta-learning evaluation for TLS, QUIC, HTTPS, IoT and IoMT traffic using leakage-aware split design and same-budget baseline controls.
- Targets anomaly and zero-day detection where labelled examples are scarce, class distributions shift and protocol behaviour changes over time.
- Directly supports the Deakin topic through robust AI for cybersecurity, privacy-preserving network monitoring and CPS-style traffic intelligence.

Data-Efficient Encrypted Traffic Intelligence | *Systematic and empirical research programme*

- Reviewed and benchmarked data-efficient learning, adaptive learning and encrypted traffic classification under modern protocols.
- Studied secure network monitoring using metadata rather than payload content, aligning with privacy-preserving cybersecurity constraints.
- Worked with IoT, IoMT, TLS, QUIC, HTTPS and VPN traffic settings relevant to AI-enabled CPS and connected infrastructure.

QKD Studio / QKD Geo Simulator | *Quantum-safe network service-management simulator* | [GitHub](#):

github.com/muntakim1/qkd-geo-simulator

A reproducible QKD/PQC network service-management simulator for key-pool dynamics, routing, scheduling, crypto-agility and closed-loop controller evaluation.

- Designed discrete-time simulation, key-pool and service-demand modelling, QKD/PQC selection, routing policies and REST/WebSocket control interfaces.
- Demonstrates capacity to build rigorous research software for critical infrastructure security, adaptive control and reproducible cyber-resilience experiments.

Crypto-Agile Quantum-Safe O-RAN Security | *Adversarial bandits for adaptive security control*

- Developed a framework for selecting quantum-safe security options under strategic eavesdropping and adversarial conditions.
- Relevant to future cyber-physical and communication systems that require adaptive security, threat modelling and trustworthy control decisions.

PUBLICATIONS AND MANUSCRIPTS

Selected publications and manuscripts below list M. Rahaman as first author unless noted otherwise.

- M. Rahaman et al., "A Systematic Literature Review on Data-Efficient and Adaptive Learning Techniques for Encrypted Traffic Classification under Modern Protocols," MDPI Computers, published.
- M. Rahaman et al., "Benchmarking Deep and Ensemble Learning for HTTPS Traffic Classification: The Case for Packet-Burst Statistics and Interpretability," IEEE TENSYP, accepted.
- M. Rahaman et al., "Evading the Strategic Eavesdropper: Adversarial Bandits for Quantum-Safe O-RAN," IEEE WIFS, in review.
- M. Rahaman et al., "QKD Studio: A Reproducible Discrete-Time Testbed for Service-Management and Closed-Loop Control Experiments in Quantum Key Distribution Networks," IEEE Transactions on Network and Service Management, in review.
- M. Rahaman et al., "Auditing Few-Shot Meta-Learning Under Domain Shift: A Control-Led Protocol for IoT and IoMT Traffic Classification," IEEE Access, ready to submit.
- M. Rahaman et al., "The Adaptation-Detection Tension in Few-Shot Traffic Classification: A Dual Readout for Zero-Day Detection," Elsevier Computer Networks, ready to submit.

TECHNICAL SKILLS

Machine learning and AI security

PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, CatBoost, few-shot learning, meta-learning, domain adaptation, drift detection, anomaly detection, zero-day detection.

Cybersecurity and CPS security

Encrypted traffic analysis, DPI/DSI, TLS, QUIC, VPN traffic, network-flow statistics, IoT and IoMT security, O-RAN security, protocol-level behaviour modelling.

Privacy-preserving monitoring

Payload-free traffic intelligence, leakage-aware evaluation, metadata-based classification, reproducible benchmarking and privacy-aware threat detection.

Cryptography and quantum-safe networking

Post-quantum cryptography concepts, QKD network service management, PQC/QKD orchestration, crypto-agility, key-as-a-service modelling and quantum-safe O-RAN security.

Software and research engineering

Python, Rust, SQL, Java, C++, C#, JavaScript, TypeScript, Git, Linux, Docker, FastAPI, Flask, REST APIs, WebSocket APIs, Tauri and React.

Data and cloud

PostgreSQL, DuckDB, BigQuery, Spark, AWS, GCP, Pandas, NumPy and Polars.

PROFESSIONAL AND APPLIED RESEARCH EXPERIENCE

Lead AI and Data Scientist / Senior Data Scientist | *BasiraTech, Innova Analytics and industry projects* | 2020 - 2025

- Built AI, data-science and analytics systems for industrial and business applications, including predictive maintenance, time-series analysis, anomaly detection, geospatial analytics and model deployment.
- Developed APIs, dashboards, reproducible workflows and stakeholder-facing technical solutions, strengthening the engineering discipline required for deployable PhD research systems.
- Translated enterprise problem definitions into measurable ML experiments, critical analysis, robust baselines and implementation-ready software.

AWARDS AND RECOGNITION

- Gold Medal, ITEX 2026, for AI-driven quantum-safe innovation involving PQC and QKD.
- Fully funded Graduate Research Assistantship, Multimedia University, supported by Telekom Malaysia.
- IEEE TENSYP accepted paper and MDPI Computers published article.
- Dean's List Award during Bachelor's studies.
- Commonwealth of Learning Scholarship, Coursera, 2020 to 2021.
- Cisco AI Innovation Award, 2025.

LANGUAGES

- English: Research writing, presentation and thesis-level academic communication. B2 or higher certificate can be provided if required.
- Bengali: Native.

REFERENCES

- Dr. Azwan Mahmud, Multimedia University, Malaysia - azwan.mahmud@mmu.edu.my
- Dr. Azlan Abd. Aziz, Multimedia University, Malaysia - azlan.abdaziz@mmu.edu.my
- Dr. Ji-Jian Chin, Plymouth University, United Kingdom - ji-jian.chin@plymouth.ac.uk